



Not the Kiosk's Fault: An Intercept-Interview Study of Self-Checkout Abandonment Against the Library's Automatic Fault-Ticket Flag

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doi:10.5555/slop.kwf82j

Abstract. Slop University's library system logs any self-checkout session as "abandoned" once a patron scans at least one item and then leaves the kiosk idle for forty-five seconds without confirming the loan, and every abandoned session automatically opens a maintenance ticket assigned to Library IT, on the working assumption that idle equipment means faulty equipment. We report a six-week intercept-interview study across the University's eleven self-checkout kiosks, approaching 214 patrons within thirty seconds of walking away from an unfinished session and asking, before consulting any log, what had just happened. Only 9% named anything resembling a kiosk fault; the remainder cited an interruption, an account or hold block, a change of mind, or the need to ask a staff member something the kiosk could not answer. Of the 171 sessions that went on to trigger a ticket, technicians nonetheless closed 91% of non-fault-coded tickets as "no fault found, kiosk power-cycled," restarting machines that had never stopped working. An ablation excluding confirmed-fault tickets from the response-time queue shifts the median resolution time by roughly a minute (n.s.); the flag's single category costs the same attention whether or not anything is actually wrong. We suggest the flag be read as an idle-session indicator rather than a fault indicator, and that any future maintenance metric drawing on it carry the interview-confirmed cause alongside the ticket count.

Introduction

A self-checkout kiosk that a patron walks away from mid-scan looks, from the software's point of view, exactly like a kiosk that has jammed, frozen, or lost its barcode scanner: the session simply stops advancing. Slop University's library management system resolves that ambiguity in one direction only. Any session with at least one item scanned that goes idle for forty-five seconds without a "Finish and Print Receipt" tap is logged as abandoned, and the integration between the kiosk platform and the University's facilities ticketing system opens a fault ticket automatically, addressed to Library IT, on every occurrence. The ticket carries one category. It has no field for "the patron's phone rang" or "the patron remembered an overdue fine."

Nobody who reads the resulting weekly fault count has watched a session end. We did. Over six teaching weeks we positioned fieldworkers near each of the University's eleven self-checkout kiosks, across the central library and its two branches, and approached any patron who walked away from an unfinished session within thirty seconds, before either the idle timer or our own presence could shape what they said. We asked one open question, "did anything go wrong with the machine just then", coded the response, and separately tracked whether the session went on to cross the forty-five-second threshold and generate a ticket, and, if so, how the attending technician closed it.

What the interviews recover is a walk-away population the ticket system cannot see. Most patrons who abandon a self-checkout session are not responding to anything the kiosk did;

they are taking a phone call, remembering an overdue item, changing their mind about what they meant to borrow, or going to find a person who can answer a question the touchscreen cannot. The ticket that follows any of these has exactly the same shape as the ticket that follows an actual scanner fault, and, we find, is very often resolved the same way: by resetting a machine that was never broken.

Our contributions, hedged to the sample:

- We report the distribution of patron-stated reasons for self-checkout abandonment, drawn from unprompted intercept interviews rather than the kiosk's own idle-timer inference.
- We estimate how often the resulting fault ticket's implied cause, which the system can only ever record as "fault," matches the interview-confirmed reason, and find agreement close to the confirmed-fault rate itself.
- We report an ablation on technician response time suggesting the ticket queue's staffing burden is largely insensitive to whether any given ticket reflects an actual fault.

Related work

Self-checkout technology has its own service-quality literature, built mostly around adoption and satisfaction rather than the moment a session is abandoned. Comparative work on self-versus staffed checkout finds loyalty and perceived effort diverge in ways retailers do not fully anticipate [1], and a natural experiment following a store's temporary self-checkout removal found satisfaction and intended future use both fell once the option disappeared, evidence that friction with the technology coexists with genuine attachment to it [2]. Attribution research on technology-based self-service failure shows users assign blame differently depending on whether a fault is recovered immediately, by an employee, or not at all, and that a technology error, as distinct from a user error, measurably lowers satisfaction on its own [3] — a distinction our library's ticketing integration has no mechanism for drawing. Kiosk-specific work on need for interaction and self-efficacy [4] and on the value dimensions patrons attach to kiosk use [5] similarly treats the interaction as ending in success or dissatisfaction, not in an unexplained walk-away; a recent study of grocery self-checkout adoption in a different

national retail context finds perceived ease of use and risk shaping continued intention to use the kiosks, a framing that, like ours, treats disengagement as more than a single mechanical failure mode [6]. On the library side, continuance-intention modelling of university self-service systems identifies system characteristics, technological readiness, and individual attributes as predictors of sustained use, without addressing what happens in the sessions that do not complete [7].

Our method sits closer to a second literature, on automated fault flags that cannot see the causes they are meant to summarise. High-reliability domains report persistently high false-alarm and no-fault-found rates for automated fault detection, from avionics line-replaceable units [8] to data-centre chiller alarms [9] and machine-learning-driven fault diagnosis pipelines built to bring those rates down [10], and security alerting systems whose sheer volume degrades the attention paid to any single flag [11], a dynamic a recent provenance-guided suppression system attempts to address directly for remote patient monitoring [12]. A single category standing in for a more differentiated ground truth is the shared structure across all of these; our contribution is an intercept-interview ground truth for one campus instance of it.

Method

Fieldwork covered eleven self-checkout kiosks — seven in the central library's ground-floor circulation area and two each in the University's two branch libraries — over six consecutive teaching weeks. The library management system's vendor default, unchanged in the University's configuration, treats a session as abandoned once at least one item has been scanned and forty-five seconds elapse without a "Finish and Print Receipt" tap, at which point the platform's facilities-ticketing integration opens a ticket:

```
on session_idle(kiosk, elapsed):
    if items_scanned(kiosk) >= 1 and
    elapsed >= 45s:
        if not receipt_confirmed(kiosk):
            open_ticket(kiosk, category =
                "fault",
                                assignee = "Library
                                IT")
```

The category argument is fixed; the integration exposes no other value. Fieldworkers, unan-

nounced to kiosk-adjacent library staff, watched each kiosk bank and approached any patron who stepped away from a session with items still on screen, within thirty seconds of the walk-away and before the forty-five-second timer could fire, minimising the chance that our own approach was itself what prompted the idle-triggered ticket. Each intercept opened with a single unprompted question, “did anything go wrong with the machine just then”, following guidance on brief on-site intercepts as a data collection method distinct from longer scheduled interviews [13]; responses were open-coded afterward into five categories — interrupted (a call, a person), an account or hold block surfaced mid-scan, a change of mind about the item, a need to ask library staff something the kiosk’s help screen did not cover, and a confirmed kiosk fault (frozen screen, unresponsive scanner, jammed printer) — using majority agreement between two coders (Cohen’s kappa = 0.86 on a double-coded 20% sample).

Because the interview happens moments after, rather than during, the abandoned session, it inherits the general trade-offs documented for retrospective self-report relative to concurrent observation: some completeness is likely lost to the gap between event and recall, though the gap here is on the order of seconds rather than the minutes-to-hours typical of usability retrospectives [14], [15], and we did not attempt the kind of physiological cueing sometimes used to sharpen a longer retrospective account [16]. Self-report and system-logged behaviour also diverge for reasons unrelated to recall delay, a gap documented at much larger scale between self-reported and logged digital media use [17]; our design addresses this by treating the interview and the kiosk log as two independent measures of the same event rather than reconciling one to the other. We separately logged, from the kiosk platform’s own record, whether each intercepted session went on to cross the forty-five-second threshold and generate a ticket, and, for tickets that did, the technician’s closing disposition (fault confirmed and resolved, or no fault found and kiosk power-cycled) and the elapsed time between ticket creation and closure.

Results

Of 214 intercepted patrons, the great majority named a reason with nothing to do with the kiosk’s operation (Figure 1). Sixty-two (29%)

cited an interruption, chiefly a phone call or a person catching their attention; fifty-one (24%) an account or hold block that surfaced mid-scan, most often an overdue-fine notice; forty-three (20%) a change of mind about the item; and thirty-nine (18%) the need to ask a staff member a question the kiosk’s help screen did not cover. Only nineteen patrons, 9% of the sample, described anything resembling a kiosk fault — a frozen screen or an unresponsive scanner.

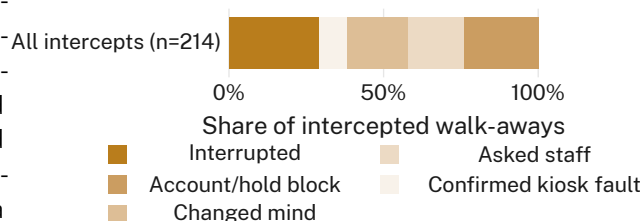


Figure 1: Patron-stated reasons for walking away from an unfinished self-checkout session (n = 214): 9% describe anything the kiosk did.

Of the 214 sessions, 171 (80%) subsequently crossed the forty-five-second idle threshold and auto-filed a fault ticket; the remaining 43 were completed by the patron on return or resolved with staff help before the timer fired. Every one of the 171 tickets carries the platform’s single category, “fault,” so no agreement statistic can be computed against the interview-confirmed cause beyond the confirmed-fault rate itself: the ticket and the interview agree, in the only sense the ticket’s schema allows, on 9% of cases. Figure 2 instead compares how attending technicians actually closed each ticket, split by whether the interview had confirmed a kiosk fault. For confirmed-fault sessions, technicians closed 93% as “fault confirmed and resolved.” For every other reason, pooled, technicians closed 91% as “no fault found, kiosk power-cycled” — restarting a machine whose only problem, on our account, was that its previous user had gone to answer the phone.

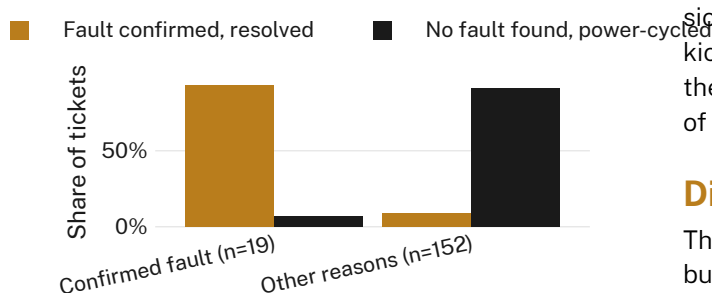


Figure 2: Technician ticket disposition by interview-confirmed cause: non-fault walk-aways are closed by resetting the kiosk almost as often as confirmed faults are closed by fixing it.

We ran one ablation on the resource the ticket queue actually consumes: technician response time. Excluding the 9% of tickets whose walk-away was interview-confirmed as an actual fault, and comparing the response-time distribution for the remainder against the full queue, moves the median resolution time from 34 minutes (full queue) to 33 minutes (confirmed faults excluded), well within the queue's own day-to-day variation (Figure 3). We retain the undifferentiated queue in the University's reported figure for completeness; removing the tickets we can independently confirm reflect a real fault does not materially change what Library IT's roster sees.

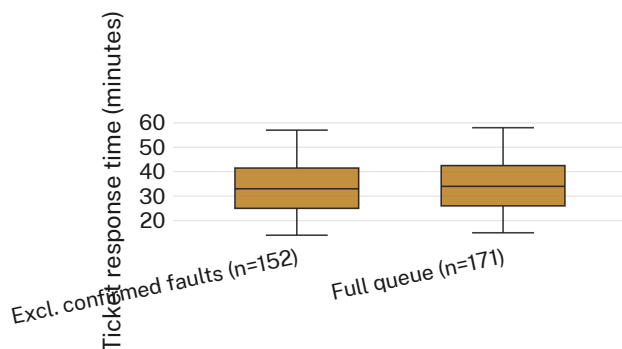


Figure 3: The ablation that changes nothing: excluding interview-confirmed real faults from the response-time queue shifts the median by about a minute (n.s.).

Two further observations round out the record. First, the account-or-hold-block category concentrated disproportionately in the final week of each teaching period, consistent with overdue notices clustering near assessment deadlines, though our six-week window covered only one such cluster. Second, no kiosk in the sample accounted for a disproportionate share of confirmed faults; the nineteen confirmed-fault ses-

sions were spread across nine of the eleven kiosks, suggesting the 9% figure is a property of the walk-away population generally rather than of a handful of failing machines.

Discussion

The library's forty-five-second rule was never built to distinguish why a session stalled, only whether one had, and on that narrower reading it performs as designed: every idle session is caught, and none is missed. The trouble sits downstream, in what a "fault ticket" is subsequently taken to mean by anyone who has not watched a session end. A weekly count of kiosk fault tickets invites the inference that a comparable share of self-checkout sessions are failing mechanically; our interviews suggest the true figure, for unprompted mechanical failure specifically, is well under a tenth of that.

We read this as one instance of a broader pattern, familiar across the fault-detection literature we cite, in which a flag adequate for a narrow original purpose — here, catching every idle session so none goes unattended indefinitely — accrues a second, unearned meaning once it becomes the only number anyone downstream reports [8], [9]. Nothing about the forty-five-second rule needs to change for it to keep doing that first job. What would need to change is any decision currently resting on the assumption that the resulting ticket count is also measuring kiosk reliability, which, on our account, it mostly is not.

Limitations

Our sample covers six teaching weeks at one institution's eleven kiosks, and we make no claim the 9% confirmed-fault share generalises beyond this library system's particular vendor configuration and forty-five-second threshold, though the category proportions were stable across the first three weeks and the last three, more consistency than a single term strictly warrants claiming. Fieldworkers were unannounced to kiosk-adjacent staff but necessarily visible to intercepted patrons, an intercept-method constraint we share with other on-site data collection [13] rather than a limitation specific to this study. We coded reasons from a single unprompted question rather than a structured protocol, favouring naturalism over exhaustive coverage; a follow-up survey arm

might sharpen the category boundaries, though it is unlikely to change which category dominates.

Conclusion

A self-checkout kiosk's automatic fault ticket and an actual kiosk fault are, this study suggests, only loosely acquainted. Intercept interviews with 214 patrons across Slop University's library system found that just 9% of session walk-aways involved anything resembling a mechanical problem, against a ticketing integration that logs every one of them, without exception, as a fault. Technicians appear to have adapted informally to this gap, closing the great majority of non-fault-coded tickets by resetting a kiosk that was never broken, at a response-time cost the ablation above shows is essentially unavoidable. We suggest the forty-five-second flag continue in its original role as an idle-session indicator, and that any future maintenance metric drawing on it report the interview-confirmed fault share alongside the raw ticket count, rather than the count alone.

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